UTM
UNIVERSITI TEKNOLOGI MALAYSIA

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I.C. No./ISID No.

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SECR1033 : COMPUTER ORGANIZATION & ARCHITECTURE

ASSIGNMENT 1.

SECTION 03.

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Question 1.

a) $-23 + (-62)$

$-23 = -(+23)$
$= -0001\ 0111$
$= 1110\ 1000 + 1$
$= 1110\ 1001$

$-62 = -(+62)$
$= -0011\ 1110$
$= 1100\ 0001 + 1$
$= 1100\ 0010$

$-23 + (-62)$:

$$\begin{array}{r} 1110\ 1001 \\ +\ 1100\ 0010 \\ \hline (1)1010\ 1011 \end{array}$$

discard

∴ No overflow occurs because negative number + negative number is still equal to a negative number. (The 8-bits result start with 1).

b) $21_{10} \times (-14)_{10} = (-21)_{10} \times 14_{10}$.

Multiplier (MP) $= 14_{10} = 001110_2$

Multiplicant (MC) $= -21_{10} = -0000\ 0001\ 0101$
$= 1111\ 1110\ 1010 + 1$
$= 1111\ 1110\ 1011$

Iteration.	Steps	Multiplier (MP)	Multiplicant (MC)	Product (P).
0	Initial value.	00 1110	1111 1110 1011	0000 0000 0000
	No Operation.			0000 0000 0000
1	Shift MC left		1111 1101 0110	
	Shift MP right	000 111		
	P=P+MC			1111 1101 0110
2	Shift MC left		1111 1010 1100	
	Shift MP right	000011		
	P=P+MC			1111 1000 0010
3	Shift MC left		1111 0101 1000	
	Shift MP right	00 0001		
	P=P+MC			1110 1101 1010
4	Shift MC left		1110 1011 0000	
	Shift MP right	00 0000		
	No Operation			1110 1101 1010
5	Shift MC left		1101 0110 0000	
	Shift MP right	00 0000		

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6	No Operation.			1110 1101 1010
	Shift MC left		1010 1100 0000	
	Shift MP right	00 0000		

$\therefore 21_{10} \times -14_{10} = 1110\ 1101\ 1010_2$

c) $13_{10} / 5_{10}$

$13_{10} = 0000001101$

$5_{10} = \quad 0101$

max bit + 1 = 5 + 1
= 6.

Iteration	Steps	Quotient(Q)	Divisor (D)	Remainder (R)
0	Initial value	0 0000	00 1010 0000	00 0000 1101
1	R=R−D			11 0110 1101
	R<0; R=D+R			00 0000 1101
	Q>Shift left(+0)	0 0000		
	D=Shift right		00 0101 0000	
2	R=R−D			11 1101 1101
	R<0; R=D+R			00 0000 1101
	Q>Shift left(+0)	0 0000		
	D=Shift right		00 0010 1000	
3	R=R−D			11 1110 0101
	R<0; R=D+R			00 0000 1101
	Q>Shift left(+0)	0 0000		
	D=Shift right		00 0001 0100	
4	R=R−D			11 1111 1001
	R<0; R=D+R			00 0000 1101
	Q>Shift left(+0)	0 0000		
	D=Shift right		00 0000 1010	
5	R=R−D			00 0000 0011
	R≥0			
	Q>Shift left(+1)	0 0001		
	D>Shift right		00 0000 0101	
6	R=R−D			11 1111 1110
	R<0; R=D+R			00 0000 0011
	Q>Shift left(+0)	0 0010		
	D>Shift right		00 0000 0010	

Quotient(Q) = $0\ 0010_2$, Remainder = $00\ 0000\ 0011_2$

$\therefore 13_{10} / 5_{10}$ = 2 remainder of 3.

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Question 2

a) $M = 1.1101_2 \times 2^6$

$N = -1.01_2 \times 2^{-10}$

① Add exponent : $6 + (-10) = -4$.

② Multiply the fraction :

$$
\begin{array}{r}
1.1101 \\
\times \quad 1.01 \\
\hline
.1\,1.1\,01 \\
00.0\,00 \\
1\,1\,1\,01 \\
\hline
10.0\,1\,0\,0\,01 \\
\hline
\end{array}
$$

$= 10.010001$

$= 10.010001 \times 2^{-4}$

③ Normalize : $10.010001 \times 2^{-4} \times 2^1 = 1.0010001 \times 2^{-3}$

④ Round the fraction (4 d.p.) : $1.0010001 \times 2^{-3} = 1.0010 \times 2^{-3}$

⑤ Set the sign : -1.0010×2^{-3}

$\therefore M \times N = -1.0010001 \times 2^{-3}$

$= -1.0010 \times 2^{-3}$.

b) Sign bit : 1 (negative number).

$2^{8-1} - 1 = 2^7 - 1 = 127$.

Biased exponent, $E_b = e' + b$

$= -3 + 127$

$= 124_{10}$

$= 0111\ 1100_2$

Mantissa / Fraction : 001 0001 → expand to 23 bits.

$= 001\ 0001\ 0000\ 0000\ 0000\ 0000$

1	0111 1100	001 0001 0000 0000 0000 0000
Sign.	Biased Exponent	Significand / Fraction.

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Question 3

(a) 0.1110111×2^2

true exponent = 2

biased value = 16

biased exponent = $16 + 2 = 18_{10} = 10010_2$

Sign	exponent	fraction
0	10010	1110 1110

$\therefore$ 0100101110 1110

(b) $13_{10} = 1101_2$

$0.625 \times 2 = 1.25 \Rightarrow 1$
$0.25 \times 2 = 0.5 \Rightarrow 0$
$0.5 \times 2 = 1.0 \Rightarrow 1$

$13.625_{10} = 1101.101 \times 2^0$
$= 0.1101101 \times 2^4$

true exponent = 4

biased value = $2^{5-1} = 16$

biased exponent = $16 + 4 = 20_{10} = 10100_2$

sign	5 bits biased exponent	8 bits fraction
1	10100	11011010

$\therefore$ 1101001101 1010

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(c) $C = 1.1011 \times 2^{-3}$ $D = -1.0001 \times 2^{-4}$

$= -0.1000 \times 2^{-3}$

$C + D = (1.1011 \times 2^{-3}) + (-0.1000 \times 2^{-3})$

$= 1.0011 \times 2^{-3}$

$$\begin{array}{r} 1.1011 \\ -0.1000 \\ \hline 1.0011 \end{array}$$

$\therefore 1.0011 \times 2^{-3}$

(d) $(2.225 \times 10^{1}) \times (1.234 \times 10^{0})$

$= 2.225 \times 1.234 \times 10^{1}$

$= 2.74565 \times 10^{1}$

$= 2.7457 \times 10^{1}$

$\therefore +2.7457 \times 10^{1}$

$$\begin{array}{r} 2.225 \\ \times\ 1.234 \\ \hline 8900 \\ 6675 \\ 4450 \\ +\ 2225 \\ \hline 2.745650 \end{array}$$